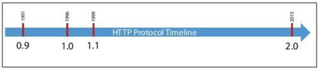

Figure 1: Pre-Internet connectivity

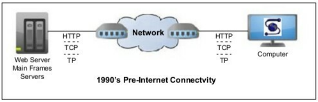

Figure 2: HTTP timeline

sections, various data types referred to in the page (namely, images), as well as audio/video links that are later fetched by the browsers to display or play. The server that serves these hypertext files is also called a HTTP (HyperText Transfer Protocol) server. This server implementation was later made into a standard by the IETF with the RFC 1945 document.

This way of exchanging documents and information became popular very quickly since the protocols that were used earlier, like the FTP (file transfer), Gopher (search and retrieve) and TELNET (a connection to another machine) were all found to be difficult to use by graduates who were not computer savvy in the college campuses. This technique caught on fast in the early 1990s.

These servers were run on mainframes and/or servers that were in the intranets of campuses and enterprises, and were primarily used to serve internal documentation and forms.

The very first website that was created by Tim Berners-Lee is still active at *http://info.cern.ch/hypertext/WWW/TheProject.html*

HTTP versions

This initial HTTP protocol version was denoted by the number 0.9. Figure 2 provides the timeline for the various versions that were released and the features that the protocols supported.

The actions that were supported in each of these versions and the standard document number (RFC) are tabulated below:

Version	Protocol actions supported	IETF RFC number(s)
0.9	GET	-
1.0	GET, POST and HEAD	1945
1.1	GET, POST, HEAD, OPTIONS, PUT, DELETE, TRACE and CONNECT	2068, 2616, 7230-7235 (superseded the previous ones)
2.0	GET, POST, HEAD, OPTIONS, PUT, DELETE, TRACE, CONNECT and PATCH	7540

The initial version supported only the GET action. This was restricted to serving the HTML files and the associated files (images) that were referred with the '<IMG>' tag. Applications at that time were mainly serving documents with images, forms that could be downloaded and printed, calendars with events marked, etc.

With the linking of email capability into the HTML syntax, there was a need to collect the feedback and contact details of people visiting the website, and sending it to appropriate personnel within the college and/or company. This is when corporate websites started appearing in the dial-up connected Internet, and these companies wanted their visitors to also fill up some forms that would enable them to collect information. This led to the POST action being supported, which would send the information collected in the browser back into the Web server for saving and sending as emails. This was the start of two-way data handling from the servers and browsers.

Browser based applications were developed, and there was a need for additional actions to be supported that would enable the server to interact with the thick client applications running on the browser to exchange data and enable workflows for the browser users. The competition amongst browser developers to capture market share did not focus on HTTP enhancements and instead laid emphasis on faster rendering of text and images in the browser itself. Since these improvements saturated, the browser vendors along with Google(SPDY) efforts started looking at how to handle the improvements via enhanced exchange of data from HTTP servers and hence this new protocol standard.

History of user agent(s) or Web browser(s)

The first browser that was developed by NCSA in 1993 was known as Mosaic and was later renamed as Netscape Navigator. Its descendant is Mozilla Firefox and, in 1995, Microsoft licensed it to create Internet Explorer. Apple released its browser, Safari, in 2003 and Google later released Chrome in 2008. Each browser competed on efficiency and faster rendering of these Hypertext files to attract a larger user base. Cross-platform (OS) versions of these browsers were also released to increase their adoption.

Apart from the performance improvements in rendering the content on the screen, efforts were made to alter content storage and retrieval to make it reach and render faster in the browsers. Pages were kept compressed and transferred to avoid latency in lower bandwidth connections. The progressive rendering of images using the interlaced GIF standard was attempted.